3.1 Introduction

herAttire Aesthetics is an AI-powered web application that helps Pakistani women try on clothes virtually before buying them. A user uploads a photo or opens her camera, selects a Pakistani outfit (shalwar kameez, kurti, abaya, lehenga, bridal wear and more), and sees a virtual try-on result. The system also estimates her body shape, approximate measurements and skin tone, and uses them to suggest the right size, colors and styles across different brands.

What makes the system unique is its Pakistani Fashion Customization module: Dulhan Mode for complete wedding looks (Mehndi, Barat, Walima), hijab and dupatta matching, and a stitching cost estimator. Users can also filter products by budget, save favorites to a wishlist, and check out using Cash on Delivery. Sellers and admins manage products, prices, size charts and orders from a separate admin dashboard.

The system has two main user types: the Customer (a Pakistani woman using the app) and the Admin/Seller (who manages the store). The system uses pretrained AI models (IDM-VTON or CatVTON for try-on, a pose-detection model for body analysis, and the Gemini AI model for styling suggestions) instead of training models from scratch.

3.3 System Features

The major features of herAttire Aesthetics are grouped below by module.

1. User & Account Management

- User registration and login
- Password reset
- Profile management
- Saved measurements, favorite colors and preferred styles
- Saved outfits
- Account deletion

2. Product Catalog & Shopping

- Product catalog with full product details
- Search, filter and sort products
- Wishlist
- Shopping cart
- Cash on Delivery (COD) checkout
- Order history and order tracking

3. Virtual Try-On

- Photo upload and camera capture
- Photo quality check
- Body pose detection
- AI-based virtual try-on using IDM-VTON or CatVTON
- View, save, download and compare try-on results
- Try-on from the product page

4. Body & Style Analysis

- Body shape estimation
- Approximate body measurement estimation
- Skin tone detection
- Review and correction of estimated values
- Saving analysis results to the profile

5. Smart Recommendation

- Smart size recommendation across different brands
- Color suggestions based on skin tone
- Style suggestions based on body shape
- Occasion-based styling
- AI mix and match within a budget
- Personal style profile that improves over time

6. Pakistani Fashion Customization

- Dulhan Mode (complete looks for Mehndi, Barat and Walima)
- Hijab and dupatta matching
- Stitching cost estimator

7. Admin Management

- Separate admin login and dashboard
- Product, brand, price, stock and size chart management
- Order management
- Stitching rate management
- Dashboard statistics (products, orders, popular items)

8. Extras innovative features

- Share try-on result with family and collect their opinion

- Live webcam try-on (optional, if time allows)

3.6 Functional Requirements

Module: User & Account Management

FE-1: User & Account Management

The system shall let a user create an account, log in, and manage her profile, saved measurements, preferences and saved outfits. This saved information is used by the recommendation module to give more personal suggestions.

FR-1.1: The system shall allow a new user to register using her name, email address, phone number and password.

FR-1.2: The system shall verify that the email address is not already registered before creating an account.

FR-1.3: The system shall validate the password strength (at least 8 characters, including a letter and a number) during registration.

FR-1.4: The system shall store user passwords in encrypted (hashed) form only.

FR-1.5: The system shall allow a registered user to log in using her email address and password.

FR-1.6: The system shall display a clear error message when the login details are incorrect or the registration details are invalid.

FR-1.7: The system shall allow a logged-in user to log out.

FR-1.8: The system shall allow a user to reset a forgotten password through a link sent to her registered email address.

FR-1.9: The system shall automatically log out a user after a period of inactivity.

FR-1.10: The system shall allow a user to view and update her profile information, such as name, phone number and profile picture.

FR-1.11: The system shall allow a user to save one or more delivery addresses.

FR-1.12: The system shall allow a user to save her body measurements, either by entering them manually or from the Body & Style Analysis result.

FR-1.13: The system shall allow a user to save her favorite colors and preferred styles.

FR-1.14: The system shall allow a user to save a try-on result or outfit to her Saved Outfits.

FR-1.15: The system shall allow a user to view and delete her Saved Outfits.

FR-1.16: The system shall allow a user to permanently delete her account and related personal data after confirmation.

FR-1.17: The system shall restrict access to a user's profile, measurements, photos and saved outfits to that user only.

FR-1.18: The system shall make the saved profile information available to the recommendation module.

Module: Product Catalog & Shopping

FE-2: Product Catalog & Shopping

The system shall provide the basic online shopping experience: browsing products, searching and filtering, wishlist, cart, Cash on Delivery checkout and order tracking.

FR-2.1: The system shall display clothing products with their name, category, brand, price, color, fabric, available sizes, occasion and images.

FR-2.2: The system shall organize products into categories such as shalwar kameez, kurti, abaya, lehenga and bridal wear.

FR-2.3: The system shall allow a user to search products by name or keyword.

FR-2.4: The system shall allow a user to filter products by category, brand, price range, color, fabric, size and occasion.

FR-2.5: The system shall allow a user to sort products by price, newest and popularity.

FR-2.6: The system shall allow a user to open a product details page that shows full product information and the brand's size chart.

FR-2.7: The system shall show the stock status of each product and size.

FR-2.8: The system shall prevent a user from adding an out-of-stock product or size to the cart.

FR-2.9: The system shall allow a user to add a product to her wishlist.

FR-2.10: The system shall allow a user to view her wishlist and remove products from it.

FR-2.11: The system shall allow a user to add a product to the cart by selecting size, color and quantity.

FR-2.12: The system shall allow a user to change the quantity of, or remove, items in the cart.

FR-2.13: The system shall display the cart items and the total price.

FR-2.14: The system shall provide a Try On option on the product details page that opens the Virtual Try-On module with that product selected.

FR-2.15: The system shall allow a user to check out using Cash on Delivery as the payment method.

FR-2.16: The system shall require a delivery address and phone number before an order can be placed.

FR-2.17: The system shall show an order summary and ask the user to confirm before placing the order.

FR-2.18: The system shall create the order and show an order confirmation with an order number after successful checkout.

FR-2.19: The system shall reduce the available stock of the ordered products when an order is placed.

FR-2.20: The system shall allow a user to view her order history and order details.

FR-2.21: The system shall show the current status of each order (Pending, Confirmed, Out for Delivery, Delivered, Cancelled).

FR-2.22: The system shall allow a user to cancel an order before it is shipped.

FR-2.23: The system shall notify the user when the status of her order changes.

Module: Virtual Try-On

FE-3: Virtual Try-On

The system shall let the user see how a selected Pakistani outfit looks on her own photo. Photo-based try-on is the guaranteed version of this module; live webcam try-on is covered separately in FE-8.

FR-3.1: The system shall allow a user to upload her photo in JPG or PNG format.

FR-3.2: The system shall allow a user to open the device camera and capture her photo directly.

FR-3.3: The system shall check that the uploaded photo is of a supported file type and within the allowed file size, and show a clear error message otherwise.

FR-3.4: The system shall check that the photo is suitable for try-on (one person, body clearly visible, enough light) and ask the user to retake it when it is not.

FR-3.5: The system shall detect the body pose of the user in the photo using a pose-detection model.

FR-3.6: The system shall allow a user to select an outfit from the product catalog for try-on.

FR-3.7: The system shall generate a virtual try-on image by placing the selected outfit on the user's photo using a pretrained AI model such as IDM-VTON or CatVTON.

FR-3.8: The system shall support Pakistani garment types, including shalwar kameez, kurti, abaya and lehenga, in the try-on.

FR-3.9: The system shall show the progress of the try-on while the result is being generated.

FR-3.10: The system shall display the try-on result next to the original photo.

FR-3.11: The system shall allow a user to try different outfits, colors or sizes on the same uploaded photo without uploading it again.

FR-3.12: The system shall allow a user to save a try-on result to her Saved Outfits.

FR-3.13: The system shall allow a user to download a try-on result.

FR-3.14: The system shall allow a user to compare two or more try-on results side by side.

FR-3.15: The system shall show an error message and allow the user to retry when the try-on result cannot be generated.

FR-3.16: The system shall inform the user that the try-on result is a visual approximation and may differ from the real outfit.

FR-3.17: The system shall allow a user to delete her uploaded photos and try-on results.

Module: Body & Style Analysis

FE-4: Body & Style Analysis

The system shall study the user's photo to estimate her body shape, approximate body measurements and skin tone. These are estimated values, not exact professional measurements, and are passed to the recommendation module.

FR-4.1: The system shall allow the user to choose whether to run the body and style analysis on her photo.

FR-4.2: The system shall estimate the user's body shape from the detected body pose points.

FR-4.3: The system shall allow the user to enter her height so that the body measurements can be estimated more accurately.

FR-4.4: The system shall estimate approximate body measurements, such as shoulder width, bust or chest, waist and hips, from the detected body pose points.

FR-4.5: The system shall detect the user's skin tone from the face area of the photo using a color-matching engine.

FR-4.6: The system shall classify the detected skin tone into a defined category (for example fair, medium, olive or dark) and undertone (warm, cool or neutral).

FR-4.7: The system shall ask the user to retake the photo when the face or body is not clearly visible for the analysis.

FR-4.8: The system shall display the analysis results clearly as estimated values with a short note that they are not exact measurements.

FR-4.9: The system shall allow the user to review and manually correct the estimated body shape, measurements and skin tone.

FR-4.10: The system shall save the analysis results to the user's profile.

FR-4.11: The system shall allow the user to run the analysis again using a new photo.

FR-4.12: The system shall pass the analysis results to the Smart Recommendation module.

Module: Smart Recommendation

FE-5: Smart Recommendation

The system shall give personal suggestions using the user's body information, preferences and the product database. It uses simple rules together with the Gemini AI model.

FR-5.1: The system shall recommend the best size for a selected product using the user's measurements and the brand's size chart.

FR-5.2: The system shall convert between the size charts of different brands so that the user can see her matching size in each brand.

FR-5.3: The system shall show the recommended size together with a short reason, for example the measurement it is based on.

FR-5.4: The system shall recommend colors that suit the user's skin tone.

FR-5.5: The system shall recommend styles that suit the user's body shape.

FR-5.6: The system shall provide occasion-based styling suggestions, such as daily wear, formal, Eid and wedding events.

FR-5.7: The system shall allow the user to select an occasion and enter a budget to get suggestions.

FR-5.8: The system shall suggest complete mix-and-match outfit combinations from the catalog that stay within the user's budget.

FR-5.9: The system shall use predefined rules for fixed logic such as size matching and color matching, and use the Gemini AI model for styling combinations and explanations.

FR-5.10: The system shall show a short explanation for each recommendation.

FR-5.11: The system shall recommend only products that are currently in stock.

FR-5.12: The system shall validate AI-generated suggestions against the product database so that only existing products are shown.

FR-5.13: The system shall build a personal style profile from the user's favorites, wishlist, saved outfits, orders and feedback.

FR-5.14: The system shall update the personal style profile as the user continues to use the system.

FR-5.15: The system shall allow the user to like or dislike a recommendation and use this feedback for later suggestions.

FR-5.16: The system shall allow the user to view and reset her style profile.

FR-5.17: The system shall show recommendations on the product details page and on a dedicated For You page.

FR-5.18: The system shall continue to provide rule-based recommendations when the Gemini AI model is not available.

Module: Pakistani Fashion Customization

FE-6: Pakistani Fashion Customization

The system shall support the needs of Pakistani women that existing apps do not cover: complete wedding looks (Dulhan Mode), hijab and dupatta matching, and an approximate stitching cost. This is the most important module of the project.

Dulhan Mode

FR-6.1: The system shall provide a Dulhan Mode for planning complete wedding looks.

FR-6.2: The system shall allow the user to select whether she is planning as a bride or as a guest.

FR-6.3: The system shall allow the user to choose the wedding events Mehndi, Barat and Walima.

FR-6.4: The system shall generate a complete look for each selected event, including the outfit, dupatta or hijab style and a matching color theme.

FR-6.5: The system shall use the user's skin tone, body shape, budget and preferences when creating the wedding looks.

FR-6.6: The system shall suggest colors and styles that are suitable for the tradition of each event.

FR-6.7: The system shall allow the user to try on each suggested wedding look virtually using the Virtual Try-On module.

FR-6.8: The system shall allow the user to view the Mehndi, Barat and Walima looks side by side for comparison.

FR-6.9: The system shall allow the user to replace any single item in a suggested look with another matching item.

FR-6.10: The system shall allow the user to save a complete wedding plan with all selected looks.

FR-6.11: The system shall display the total estimated cost of all selected wedding looks.

Hijab & Dupatta Matching

FR-6.12: The system shall suggest matching hijab or dupatta colors and styles for a selected outfit.

FR-6.13: The system shall support hijab and dupatta suggestions for outfits such as abayas, shalwar kameez and lehengas.

FR-6.14: The system shall display the suggested hijab or dupatta together with the selected outfit so that the user can compare them.

FR-6.15: The system shall allow the user to choose from more than one matching suggestion.

Stitching Cost Estimator

FR-6.16: The system shall provide a stitching cost estimator that gives an approximate tailoring price.

FR-6.17: The system shall allow the user to enter the garment type, fabric, embellishment level (plain, light or heavy), lining and number of pieces for the estimate.

FR-6.18: The system shall calculate the estimate using stitching rates that the admin can update.

FR-6.19: The system shall display a breakdown of the estimated stitching cost.

FR-6.20: The system shall inform the user that the stitching cost is an approximate value and the final price may differ.

FR-6.21: The system shall add the estimated stitching cost to the total cost of a wedding plan when the user chooses to.

Module: Admin Management

FE-7: Admin Management

The system shall give the seller or admin a separate dashboard to manage the products, prices, size charts, orders and other store data of the platform.

FR-7.1: The system shall provide a separate login for the admin.

FR-7.2: The system shall allow only authorized admin users to access the admin dashboard.

FR-7.3: The system shall allow the admin to add a new product with its name, category, brand, price, color, fabric, sizes, occasion, stock and images.

FR-7.4: The system shall allow the admin to edit the details of an existing product.

FR-7.5: The system shall allow the admin to remove a product from the catalog.

FR-7.6: The system shall prevent permanent deletion of a product that has active orders, and allow the admin to deactivate it instead.

FR-7.7: The system shall ask the admin for confirmation before removing a product or brand.

FR-7.8: The system shall allow the admin to add, edit and remove brands.

FR-7.9: The system shall allow the admin to update the prices of products.

FR-7.10: The system shall allow the admin to add and update the size chart of each brand.

FR-7.11: The system shall allow the admin to update the stock quantity of each product and size.

FR-7.12: The system shall allow the admin to view all customer orders and their details.

FR-7.13: The system shall allow the admin to filter orders by status and date.

FR-7.14: The system shall allow the admin to update the status of an order.

FR-7.15: The system shall allow the admin to manage the stitching rates used by the stitching cost estimator.

FR-7.16: The system shall display basic statistics on the dashboard, including the number of products, the number of orders and the popular items.

FR-7.17: The system shall determine popular items using data such as orders, wishlist additions and try-ons.

FR-7.18: The system shall record the important actions performed by the admin, such as product and order changes.

Module: Innovative Extras: Share With Family and Live Try-On

FE-8: Innovative Extras: Share With Family and Live Try-On

The system shall help users decide on expensive outfits by sharing results with family, and shall offer live try-on as an optional extra if time allows.

FR-8.1: The system shall allow a user to create a private shareable link for a selected try-on result.

FR-8.2: The system shall show only the shared try-on result and product details on the shared link, and not the user's profile, measurements or other photos.

FR-8.3: The system shall allow a person who opens the shared link to give a simple opinion, such as like, dislike or a short comment.

FR-8.4: The system shall show the opinions received on a shared try-on result to the user.

FR-8.5: The system shall allow the user to turn off a shared link at any time.

FR-8.6: The system shall allow a user to try on an outfit live through the device webcam. (Optional, only if time allows.)

FR-8.7: The system shall fall back to the photo-based try-on when the webcam or live try-on is not available. (Optional, only if time allows.)

3.7 Non-Functional Requirements

Non-functional requirements describe how well herAttire Aesthetics should work. They define the expected functionality, usability, performance, reliability, maintainability, portability, compatibility and security of the system. The numeric targets are measurable so they can be tested.

NFR-1: Functionality – Suitability

The system shall provide features that are suitable for online clothes shopping by Pakistani women. It shall support Pakistani garment types, wedding events (Mehndi, Barat, Walima) and hijab/dupatta matching, and shall give suggestions that are relevant to the user's body information, preferences and budget.

NFR-2: Functionality – Accuracy

The size recommendation shall follow the brand's size chart exactly when it is rule-based. Estimated body measurements shall be shown as approximate values and shall normally be within

about 5 cm of manual measurements during testing. The try-on result shall place the selected outfit correctly on the user's body in most test photos, and the user shall be told that results are visual approximations.

NFR-3: Usability – Learnability and Understandability

The system shall have a simple and easy-to-use web interface in clear language. A first-time user shall be able to upload a photo and see a try-on result within 3 minutes without help, and at least 90% of test users shall complete this task successfully. Important actions such as Try On, Add to Cart and Save Outfit shall be reachable within 3 interactions.

NFR-4: Usability – Accessibility

The system shall use readable text, sufficient color contrast and alternative text for important images and icons. Text shall be resizable up to 200% without hiding or overlapping important controls, and color shall not be the only way to show important information such as stock status or order status.

NFR-5: Efficiency – Time Behavior

Normal pages shall load within 3 seconds. Body and style analysis shall normally finish within 15 seconds, recommendations within 5 seconds, and a try-on result within 60 seconds on the GPU/cloud server. The system shall show progress while the try-on or analysis is running so that the user knows processing is in progress.

NFR-6: Efficiency – Capacity

The system shall support at least 50 users using the system at the same time without failure. When many try-on requests arrive together, the system shall place them in a queue and show the user her waiting status instead of failing.

NFR-7: Reliability – Fault Tolerance and Availability

If an AI service (try-on model, pose detection or Gemini) is not available, the system shall continue to provide the shopping features such as browsing, wishlist, cart and checkout. When the Gemini AI model is not available, the system shall continue to give rule-based recommendations. The system shall show a clear message instead of a technical error.

NFR-8: Reliability – Recoverability

The system shall not lose placed orders, saved outfits, profile data or wedding plans when a try-on or analysis request fails or the application has an unexpected error. The user shall be able to retry a failed request, and the system database shall be backed up regularly.

NFR-9: Maintainability – Modularity

The system shall be built from separate modules: Account, Catalog & Shopping, Virtual Try-On, Body & Style Analysis, Recommendation, Pakistani Fashion Customization and Admin. Each module shall be updated or fixed without unnecessarily affecting the others. The try-on model (for example IDM-VTON or CatVTON) shall be replaceable without changing the other modules.

NFR-10: Portability – Adaptability

The system shall work on major modern web browsers, including Chrome, Firefox, Edge and Safari. The interface shall be responsive and adjust to desktop, laptop, tablet and mobile screens, and the camera capture shall work on mobile browsers where the device supports it.

NFR-11: Compatibility – Interoperability

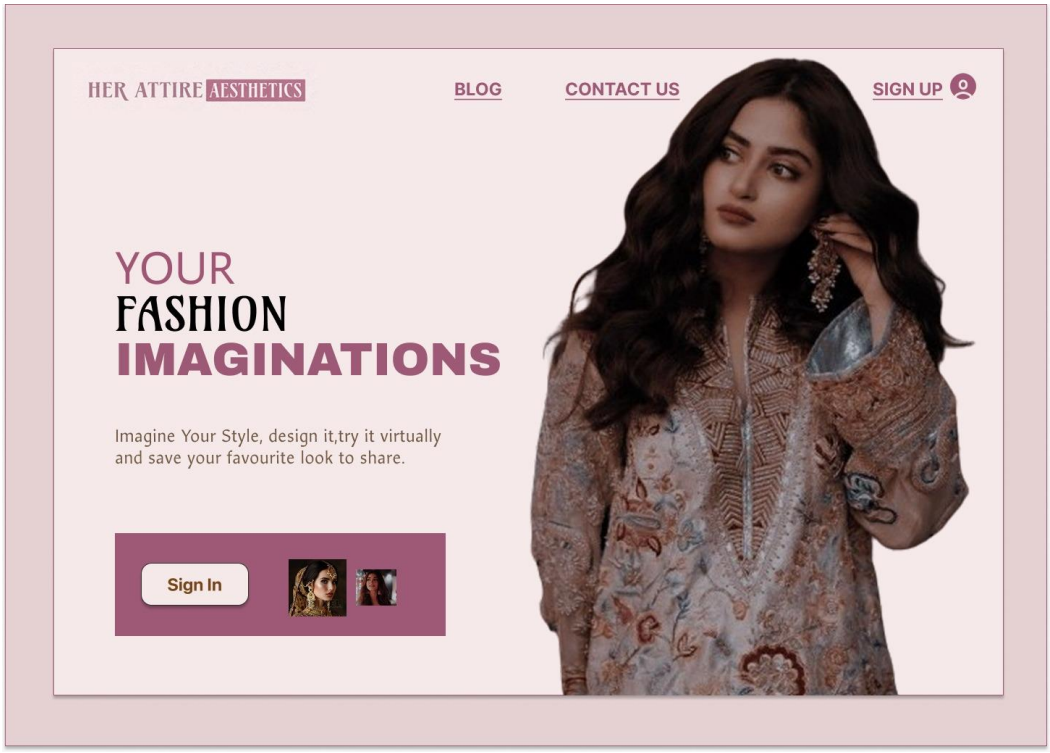
The system shall connect with the pretrained try-on model, the pose-detection model and the Gemini AI model through defined interfaces. The system shall send only the data needed for the current request to these services and shall handle their failures or timeouts properly.

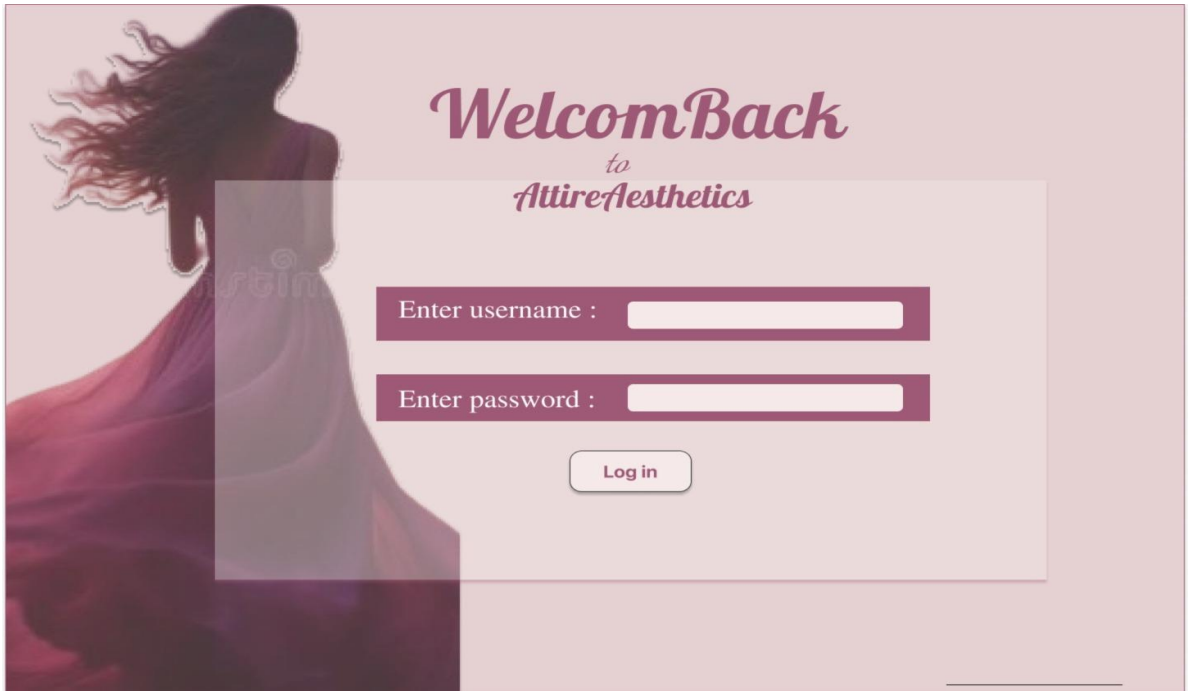
NFR-12: Security – Confidentiality and Access Control

The system shall protect user and admin data from unauthorized access. Communication between the user and the system shall use secure connections (HTTPS), passwords shall be stored in hashed form, and access to customer and admin functions shall be controlled by role. All user inputs and uploaded files shall be validated to protect against common attacks such as SQL injection and cross-site scripting. Passwords, API keys and other secrets shall not appear in source code or error messages.

NFR-13: Security – Privacy of Photos and Personal Data

User photos, try-on results, body measurements and skin tone are private data. The system shall ask for the user's permission before processing her photo, make these data visible only to the owner, and allow the user to delete her photos, results and account data at any time. Photos shall not be shared with other users, except through a link that the user herself creates, and shall not be used for any purpose other than the features of the system.


HER ATTIRE AESTHETICS
BLOG
CONTACT US
SIGN UP
YOUR
FASHION
IMAGINATIONS
Imagine Your Style, design it,try it virtually
and save your favourite look to share.
Sign In


WelcomBack
to
AttireAesthetics
Enter username :
Enter password :
Log in

Create Account
To
Create your Looks

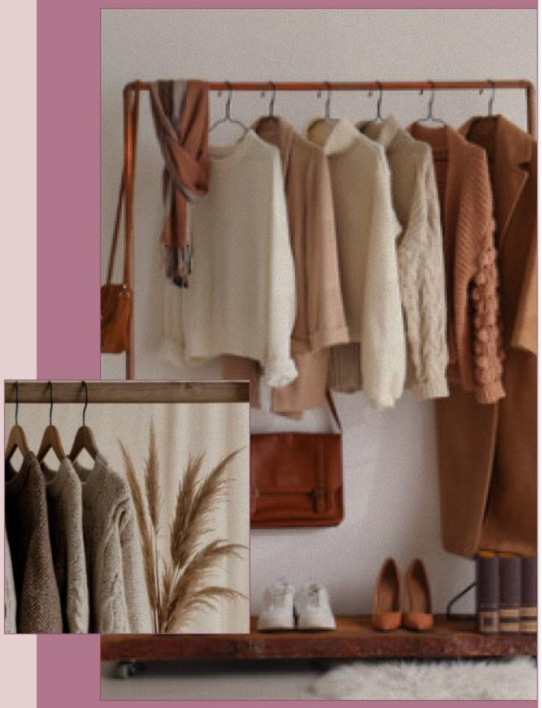
Enter username/email: herattire@gmail.com

Enter password : smhut#@!76565m

Confirm password: smhut#@!76565m

Sign Up

Already have account?Sign in



Dashboard

Her Attire

- Home
- Shop
- Live Me
- For You
- Personalized Style
- My Looks
- Orders
- Dulhan Mode
- Profile

Log Out

Search Outfits styles

Hello, Sana

Live Me
upload photo/use camera
Try Now

My Looks
Outfits you saved
View

Personlized Style
Suggestions for you
Get

Dulhan Mode
Mehndi,barat,walima
Plan Looks

HER ATTIRE AESTHETICS

Forgot your password?

Enter your email address and
send you reset link

EMAIL ADDRESS

Enter your email address

SEND RESET LINK →

← Back to Login





Explore brands

of your own choice to get their latest
colllection and others.



Live me

Choose how you want to try your outfit



Explore Brands



Choose from Gallery





Design Dress



Retake Photo

Save Photo

Don't Save